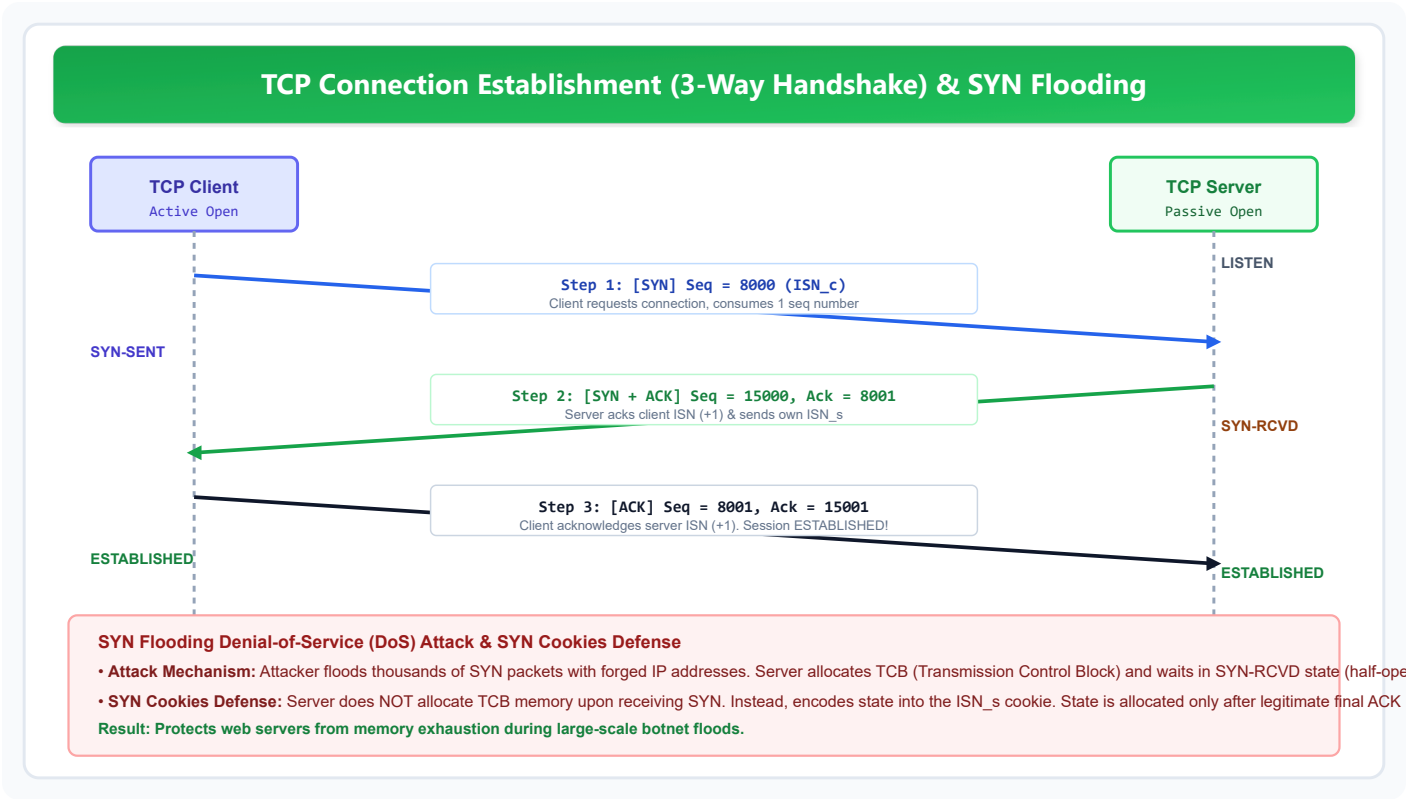


Module 03: TCP Connection Establishment and SYN Flooding

Unit 4: Transport Layer

3-Way Handshake & Security

TCP Three-Way Handshake Sequence, State Transitions, Sequence Consumption Rules, SYN Flooding DoS Attacks, and Cryptographic SYN Cookies Defense.



1. 3-Way Handshake Sequence Breakdown

Step	Sender -> Receiver	Active Flags	Seq / Ack Equations	State Transition
1	Client -> Server	SYN = 1	Seq = ISN_c (consumes 1 seq)	Client: CLOSED -> SYN-SENT

2	Server -> Client	SYN = 1, ACK = 1	Seq = ISN _s , Ack = ISN _c + 1	Server: LISTEN -> SYN-RCVD
3	Client -> Server	ACK = 1	Seq = ISN _c + 1, Ack = ISN _s + 1	Both enter ESTABLISHED state

2. SYN Cookies Defense Principle

Zero State Allocation: Server does not allocate memory on SYN arrival.

$ISN_s = \text{HMAC}(\text{Client_IP}, \text{Client_Port}, \text{Server_IP}, \text{Server_Port}, \text{Counter}, \text{Key})$

State is instantiated only upon reception of verified final ACK!